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Ans: C

$$\langle P \rangle = V_{rms} I_{rms}$$

$$(I_{rms})_{min} = 2000 / 240 = 8.33 \text{ A}$$

$$(I_0)_{min} = 8.33 \times \sqrt{2} = 11.8 \text{ A}$$

$$(I_{rms})_{max} = 2400 / 220 = 10.9 \text{ A}$$

$$(I_0)_{max} = 10.9 \times \sqrt{2} = 15.4 \text{ A}$$

$$\omega = 2\pi f$$

$$\omega_{min} = 2\pi \times 50 = 314$$

$$\omega_{max} = 2\pi \times 60 = 377$$

Probable current, $I = 14.1 \sin(375t)$

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Ans: B